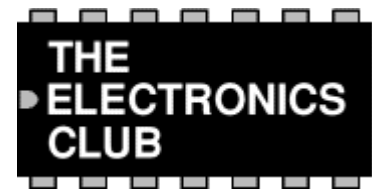


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Impedance and Reactance



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Impedance

Impedance (symbol Z) is a measure of the overall opposition of a circuit to current, in other words: how much the circuit **impedes** the flow of current. It is like resistance, but it also takes into account the effects of capacitance and inductance. Impedance is measured in ohms, symbol Ω .

Impedance is more complex than resistance because the effects of capacitance and inductance vary with the frequency of the current passing through the circuit and this means **impedance varies with frequency**! The effect of resistance is constant regardless of frequency.

The term 'impedance' is often used (quite correctly) for simple circuits which have no capacitance or inductance - for example to refer to their '[input impedance](#)' or '[output impedance](#)'. This can seem confusing if you are learning electronics, but for these simple circuits you can assume that it is just another word for resistance.

$$\text{Impedance, } Z = \frac{V}{I}$$

$$\text{Resistance, } R = \frac{V}{I}$$

V = voltage in volts (V)
 I = current in amps (A)
 Z = impedance in ohms (Ω)
 R = resistance in ohms (Ω)

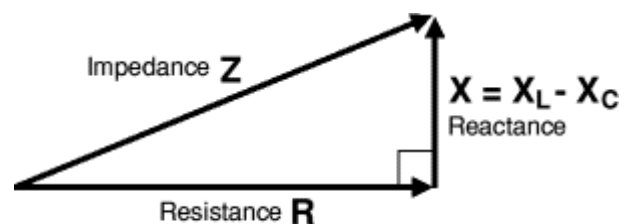
Four electrical quantities determine the impedance (Z) of a circuit: [resistance](#) (R), [capacitance](#) (C), inductance (L) and [frequency](#) (f).

Impedance can be split into two parts:

- **Resistance R** (the part which is constant regardless of frequency)
- **Reactance X** (the part which varies with frequency due to capacitance and inductance)

For further details please see the section on [Reactance](#) below.

The capacitance and inductance cause a **phase shift*** between the current and voltage which means that the resistance and reactance cannot be simply added up to give impedance. Instead they must be **added as vectors** with reactance at right angles to resistance as shown in the diagram.



$$\text{Impedance, } Z = \sqrt{R^2 + X^2}$$

* **Phase shift** means that the current and voltage are out of step with each other. Think of [charging a capacitor](#). When the voltage across the

capacitor is zero, the current is at a maximum; when the capacitor has charged and the voltage is at a maximum, the current is at a minimum. The charging and discharging occur continually with AC and the current reaches its maximum shortly before the voltage reaches its maximum: so we say the current leads the voltage.

Reactance, X

Reactance (symbol X) is a measure of the opposition of capacitance and inductance to current. Reactance varies with the frequency of the electrical signal. Reactance is measured in ohms, symbol Ω .

There are two types of reactance: capacitive reactance (X_C) and inductive reactance (X_L).

The **total reactance (X)** is the *difference* between the two: $X = X_L - X_C$

- **Capacitive reactance, X_C**

$$X_C = \frac{1}{2\pi f C} \quad \text{where: } \begin{array}{l} X_C = \text{reactance in ohms } (\Omega) \\ f = \text{frequency in hertz (Hz)} \\ C = \text{capacitance in farads (F)} \end{array}$$

X_C is large at low frequencies and small at high frequencies.
For steady DC which is zero frequency, X_C is infinite (total opposition),
hence the rule that **capacitors pass AC but block DC**.

For example: a $1\mu\text{F}$ capacitor has a reactance of $3.2\text{k}\Omega$ for a 50Hz signal,
but when the frequency is higher at 10kHz its reactance is only 16Ω .

- **Inductive reactance, X_L**

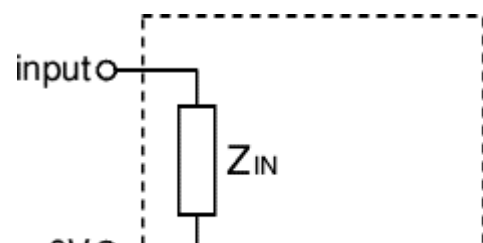
$$X_L = 2\pi f L \quad \text{where: } \begin{array}{l} X_L = \text{reactance in ohms } (\Omega) \\ f = \text{frequency in hertz (Hz)} \\ L = \text{inductance in henrys (H)} \end{array}$$

X_L is small at low frequencies and large at high frequencies.
For steady DC (frequency zero), X_L is zero (no opposition),
hence the rule that **inductors pass DC but block high frequency AC**.

For example: a 1mH inductor has a reactance of only 0.3Ω for a 50Hz signal,
but when the frequency is higher at 10kHz its reactance is 63Ω .

Input Impedance Z_{IN}

Input impedance (Z_{IN}) is the impedance 'seen' by anything connected to the input of a circuit or device (such as an amplifier). It is the combined effect of all the resistance, capacitance and inductance connected to the input inside the circuit or device.



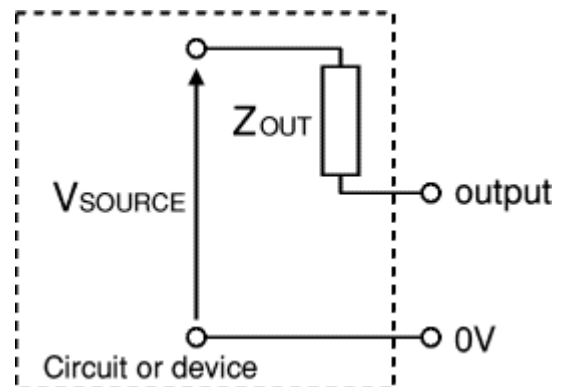
It is normal to use the term 'input impedance' even for simple cases where there is only resistance and the term 'input resistance' could be used instead. In fact it is usually reasonable to assume that an input impedance is just resistance providing the input signal has a low frequency (less than 1kHz say).

The effects of capacitance and inductance vary with frequency, so if these are present the input impedance will vary with frequency. The effects of capacitance and inductance are generally most significant at high frequencies.

Usually **input impedances should be high**, at least ten times the output impedance of the circuit (or component) supplying a signal to the input. This ensures that the input will not 'overload' the source of the signal and reduce the strength (voltage) of the signal by a substantial amount.

Output Impedance Z_{OUT}

The output of any circuit or device is equivalent to an output impedance (Z_{OUT}) in series with a perfect voltage source (V_{SOURCE}). This is called the **equivalent circuit** and it represents the combined effect of all the voltage sources, resistance, capacitance and inductance connected to the output inside the circuit or device. Note that V_{SOURCE} is usually **not** the same as the supply voltage V_s .



The equivalent circuit of any output

It is normal to use the term 'output impedance' even for simple cases where there is only resistance and the term 'output resistance' could be used instead. In fact it is usually reasonable to assume that an output impedance is just resistance providing the output signal has a low frequency (less than 1kHz say).

The effects of capacitance and inductance vary with frequency, so if these are present the output impedance will vary with frequency. The effects of capacitance and inductance are generally most significant at high frequencies.

Usually **output impedances should be low**, less than a tenth of the load impedance connected to the output. If an output impedance is too high it will be unable to supply a sufficiently strong signal to the load because most of the signal's voltage will be 'lost' inside the circuit driving current through the output impedance Z_{OUT} . The load could be a single component or the input impedance of another circuit.

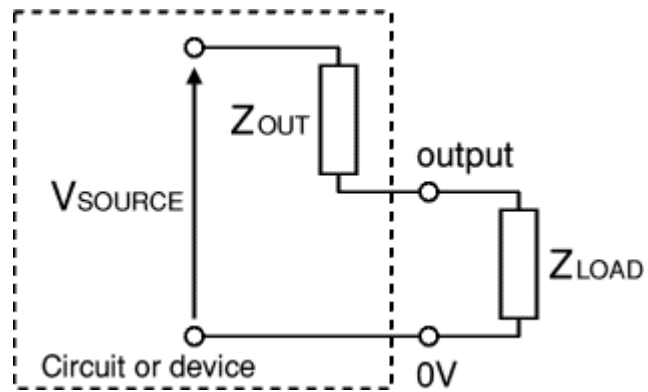
- **Low output impedance**,
 $Z_{OUT} \ll Z_{LOAD}$
 Most of V_{SOURCE} appears across the load, very little voltage is 'lost' driving the output current through the output impedance. Usually this is the best arrangement.
- **Matched impedances**, $Z_{OUT} = Z_{LOAD}$

Half of V_{SOURCE} appears across the load, the other half is 'lost' driving the output current through the output impedance. This arrangement is useful in some situations (such as an amplifier driving a loudspeaker) because it delivers **maximum power to the load**. Note that an equal amount of power is wasted driving the output current through Z_{OUT} , an efficiency of 50%.

- **High output impedance,**

$$Z_{\text{OUT}} \gg Z_{\text{LOAD}}$$

Only a small portion of appears across the load, most is 'lost' driving the output current through the output impedance. This arrangement is unsatisfactory.



The load can be a single component or the input impedance of another circuit

The output resistance of a voltage divider

[Voltage dividers](#) are widely used in electronics, for example to connect an input transducer such as an LDR to a circuit input.

For successful use the output impedance of the voltage divider should be much smaller than the input impedance of the circuit it is connected to. Ideally the output impedance should be less than a tenth of the input impedance.

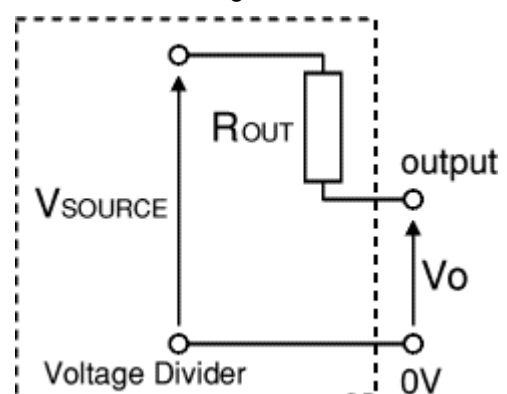
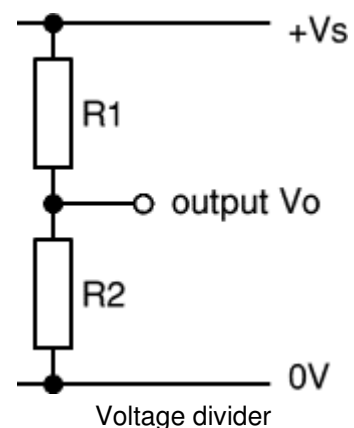
In the equivalent circuit of a voltage divider the output impedance is just a resistance and the term 'output resistance' could be used. R_{OUT} is equal to the two resistances (R_1 and R_2) connected in [parallel](#):

$$\text{Output impedance, } R_{\text{OUT}} = \frac{R_1 \times R_2}{R_1 + R_2}$$

The voltage source V_{SOURCE} in the equivalent circuit is the value of the output voltage V_o when there is nothing connected to the output (and therefore no output current). It is sometimes called the 'open circuit' voltage.

$$\text{Voltage source, } V_{\text{SOURCE}} = \frac{V_s \times R_2}{R_1 + R_2}$$

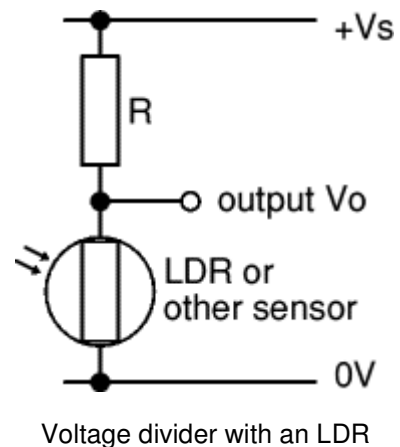
In most voltage dividers one of the resistors will be an



Equivalent circuit of a voltage divider

[input transducer](#) such as an LDR. The transducer's resistance varies and this will make both V_{SOURCE} and R_{OUT} vary too. To check that R_{OUT} is sufficiently low you should work out its highest value which will occur when the transducer has its maximum resistance (this applies wherever the transducer is connected in the voltage divider).

For example: If $R_1 = 10\text{k}\Omega$ and R_2 is an LDR with maximum resistance $1\text{M}\Omega$,
 $R_{\text{OUT}} = 10\text{k} \times 1\text{M} / (10\text{k} + 1\text{M}) = 9.9\text{k}\Omega$ (say $10\text{k}\Omega$).
 This means it should be connected to a load or input resistance of at least $100\text{k}\Omega$.



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